Animatronic Hand

BY GONE RAMU

Project Aim

- To make an Animatronic Hand using Flex Sensor and Arduino Mega.

Apparatus Required

Hardware Tools :

1. Arduino Mega
2. 5 Flex Sensor (Aluminium , carbon polymer , jumper wires, insulation tape)
3. 5 Servo Motors
4. Corogated Plastic
5. Breadboard
6. 5 Resistors
7. Hand Gloves
8. Jumper wires
9. Arduino & USB cable
10. 2 Battery Source
11. 5 Bolts
12. Threads & rubbers

Apparatus Required

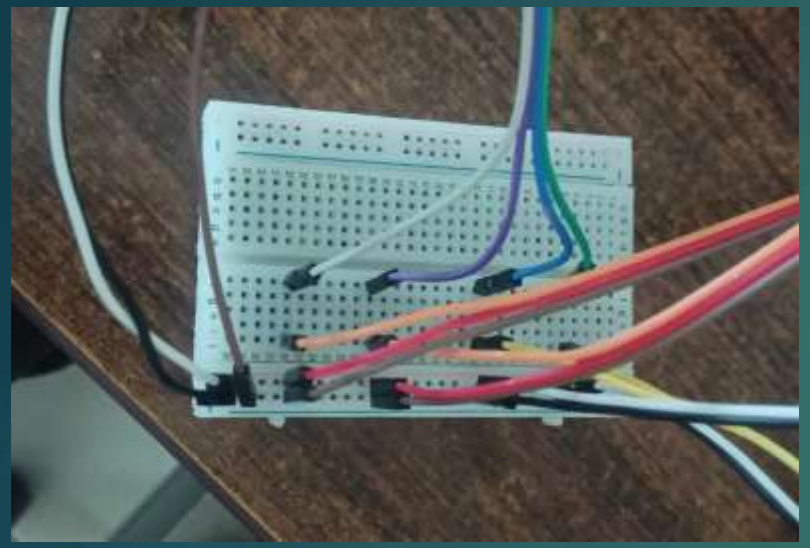
Software Tools :

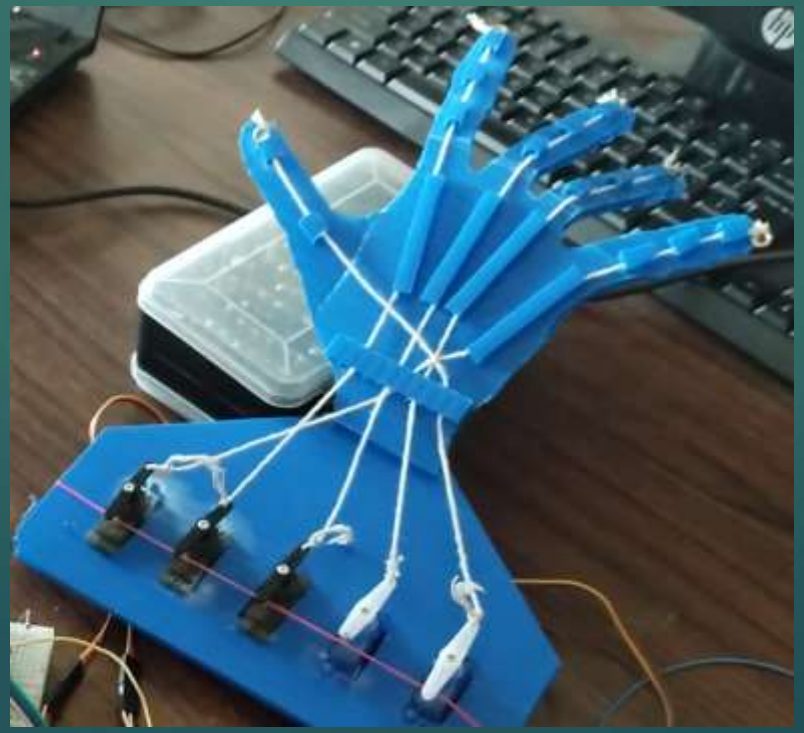
1. Arduino IDE

Procedure

- Cut the coro plast according to your left hand . Fix the rubbers and threads for each finger accordingly.
- Cut the coro plast again for servo motors placement.
- Prepare Flex sensor using aluminum, insulation tape, jumper wires, carbon polymer.
- Fix the 5 flex sensors on your 5 fingers. Integrate them with Arduino mega
- Place the 5 servo motors on coro plast .Connect them with threads of each respective finger. Integrate servos with Arduino mega.
- Finally upload the project code in Arduino mega.

Circuit Connection





Servos connection with Batteries :

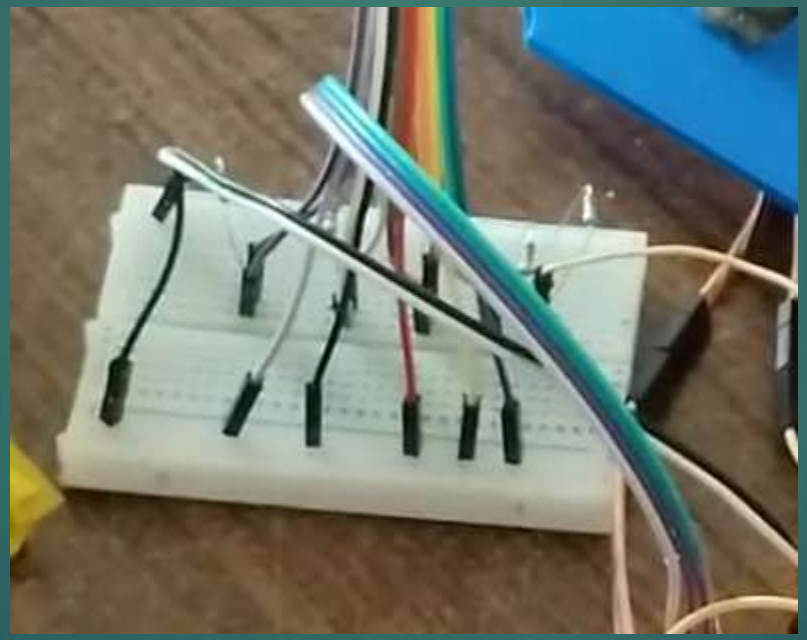
4 Servos	Battery
VCC	Anode
GND	Cathode/GND

Servos connection with Arduino mega :

Servo signal pin	Arduino mega
Servo1	12
Servo2	9
Servo3	10
Servo4	11
Servo5	8

Circuit Connection

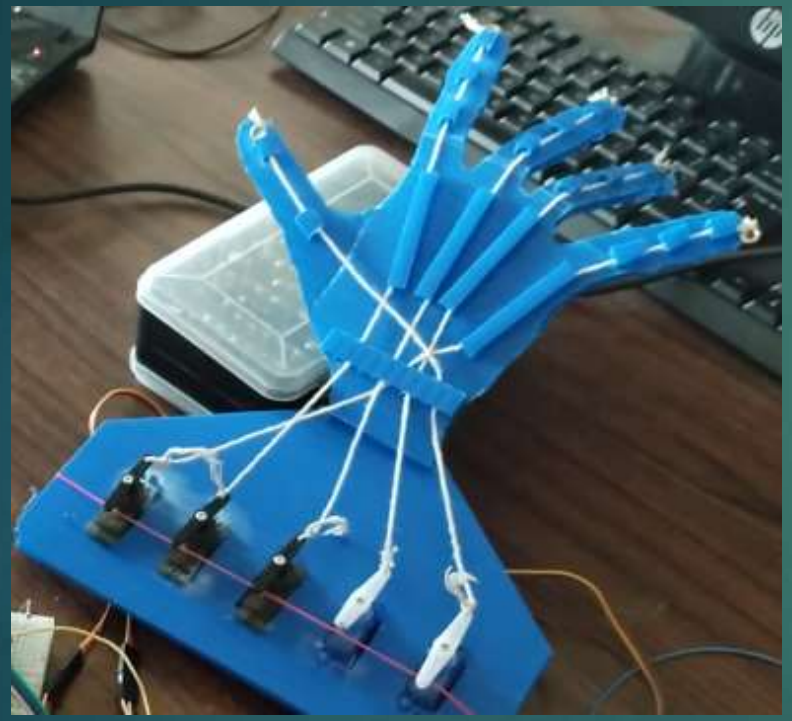




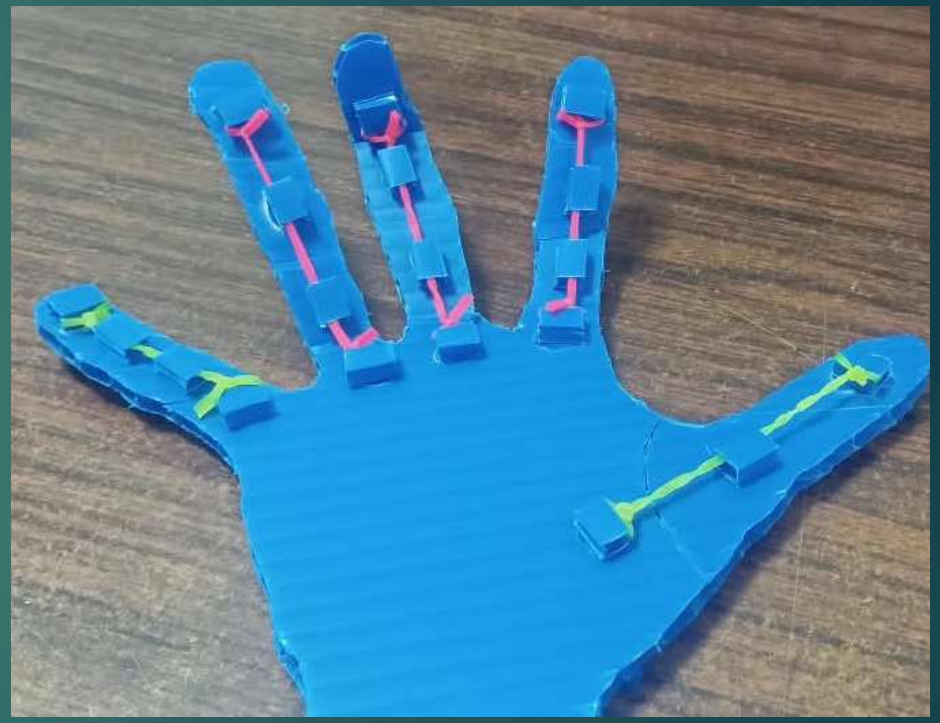
Flex Sensors connection with Arduino mega :

Flex Sensor	Arduino mega
A	A4
B	A3
C	A2
D	A1
E	A0

Animatronic Hand Preparation



Animatronic Hand one side view



Animatronic Hand other side view

Result

- Finally, we made Animatronic Hand which mimics 5 fingers of Human Hand using Flex sensors and Arduino Mega.

